

SnehoAyu

স্নেহ আয়ু

Step 1 — Onboarding Flow & Database Schema

Foundation document before development begins

Version 1.0 | June 2025

Prepared for: Development Team

This document covers: (A) Complete Login & Signup Flow with Language Selection | (B) Full Database Schema for all 4 system entities

Part A — Onboarding & Login Flow

This section describes every screen a mother or nurse sees from the moment they open the app for the first time, all the way to landing on the Home Dashboard. Developers should treat this as the exact specification for building these screens.

A.1 First Launch — Language Selection Screen

Screen: /language-select | Who sees it: Everyone on first open | Required: Yes

This is the very first screen when the app is opened for the first time on any phone. Before anything else, the user picks their language. This setting is saved to the device and used for all app text going forward.

What the Screen Shows

- SnehoAyu logo and Bengali name স্নেহ আয়ু centred at top
- Tagline in all 3 languages: 'Your baby's care companion' / 'আপনার শিশুর যত্নের সঙ্গী' / 'आपके शिशु की देखभाल का साथी'
- 3 large tappable language buttons — one per language

Button	Language Label	Sub-label	Stores As
Button 1	বাংলা	Bengali	lang = 'bn'
Button 2	हिंदी	Hindi	lang = 'hi'
Button 3	English	English	lang = 'en'

Behaviour

- User taps a language — it is saved to localStorage as preferred_language
- On ALL future app opens, language selection is skipped — app opens in saved language
- Language can be changed later from Profile → Settings
- After selecting language — user is taken to the Welcome / Splash screen

Developer note: Use react-i18next. All UI strings must have keys in bn.json, hi.json, en.json. Bengali (bn) is the default if no language is saved.

A.2 Welcome Screen

Screen: /welcome | Who sees it: First-time users after language select

What the Screen Shows

- SnehoAyu logo
- Headline (in selected language): 'Supporting you and your baby after NICU discharge'
- 3 small feature highlights with icons: Daily Care Guide | Growth Tracking | Danger Signs

- Two buttons: 'Create Account' (primary, filled) and 'I Already Have an Account' (secondary, outlined)

Behaviour

- 'Create Account' → goes to A.3 Phone Number Entry (signup)
- 'I Already Have an Account' → goes to A.6 Returning Login
- This screen is NEVER shown again after the first login — returning users go directly to login

A.3 Signup — Step 1: Phone Number Entry

Screen: /signup/phone | Who sees it: New mothers only

What the Screen Shows

- Heading: 'Enter your mobile number'
- Subtext: 'We will send you a verification code by SMS'
- Phone number input field — numeric keyboard, auto-prefixes +91 (India)
- 'Send OTP' button — disabled until 10 digits entered
- Small text: 'Your number will only be used for login and study communication'

Validation

Rule	Behaviour
Must be exactly 10 digits	Show inline error: 'Please enter a valid 10-digit mobile number'
Number already registered	Show: 'This number already has an account. Did you mean to log in?' + link to login
SMS send fails	Show: 'Could not send SMS. Please check your number or try again.'

On 'Send OTP' tap

- API call: POST /auth/send-otp { phone: '9XXXXXXXXX' }
- Backend sends 6-digit OTP via SMS (MSG91 gateway)
- OTP is valid for 10 minutes
- User is navigated to A.4 OTP Verification screen
- Phone number is passed forward as state — not re-entered

A.4 Signup — Step 2: OTP Verification

Screen: /signup/verify-otp | Who sees it: After phone number entry

What the Screen Shows

- Heading: 'Enter the 6-digit code'

- Subtext: 'Sent to +91 XXXXXX1234' (last 4 digits shown)
- 6 individual digit input boxes (OTP style) — auto-advances focus
- Countdown timer: 'Resend in 02:34'
- 'Resend OTP' link — active after 3 minutes
- 'Verify' button

Behaviour

Scenario	Behaviour
Correct OTP entered	Navigate to A.5 PIN creation
Wrong OTP	Show: 'Incorrect code. X attempts remaining.' (3 attempts max)
3 wrong attempts	Lock for 30 minutes. Show countdown.
OTP expired (>10 min)	Show: 'Code has expired. Please request a new one.'
Resend OTP tapped	Re-send OTP. Reset countdown. Show: 'New code sent.'

A.5 Signup — Step 3: Create 4-Digit PIN

Screen: /signup/create-pin | Who sees it: After OTP verified

A 4-digit PIN is used instead of a password because mothers may have low literacy or small screens. It is fast to enter and easy to remember.

What the Screen Shows

- Heading: 'Create a 4-digit PIN'
- Subtext: 'You will use this to open the app'
- 4 PIN input dots (not digits shown — masked)
- Numeric keypad on screen
- After entering 4 digits: screen changes to 'Confirm your PIN'
- User enters PIN again to confirm
- 'Set PIN' button appears after confirmation entry

Validation

- If confirm PIN does not match: 'PINs do not match. Please try again.' — clears both fields
- Do not allow: 0000, 1234, 1111 (too simple — show warning but allow user to proceed)
- PIN is hashed (bcrypt) before storing — never stored in plain text

A.6 Returning User — Login Screen

Screen: /login | Who sees it: Mothers who already have an account

What the Screen Shows

- SnehoAyu logo (small, top)
- Heading: 'Welcome back'
- Phone number input (same as A.3)
- 'Continue' button
- On phone verified: shows 4-dot PIN entry with on-screen numeric keypad
- 'Forgot PIN?' link below keypad

Login Flow

1. User enters phone number → taps Continue
2. Backend checks if phone exists → if yes, show PIN screen
3. User enters 4-digit PIN
4. Backend verifies PIN hash → issues JWT access token (24 hours) + refresh token (30 days)
5. User navigated to Home Dashboard
6. Session stays active 30 days — user does not log in again unless they log out manually

Forgot PIN Flow

7. User taps 'Forgot PIN?'
8. OTP sent to registered phone number
9. User verifies OTP → goes to A.5 PIN creation flow to set new PIN
10. Old PIN invalidated immediately

Security note: After 5 wrong PIN attempts, account is locked for 30 minutes. Show a countdown. Send SMS alert to the phone number: 'Someone tried to log into your SnehoAyu account.'

A.7 Signup — Step 4: Mother Profile Form

Screen: /signup/mother-profile | Who fills it: Nurse + Mother together at hospital

This screen collects all demographic data for Tool I of the research proposal. It is filled once during hospital onboarding. The nurse sits with the mother and enters this together. All fields are required unless marked optional.

Form Fields — Mother Information

Field	Type	Options / Validation	Required
Parent's Name	Text input	Free text. Note: stored separately from participant code for privacy	Optional
Age	Dropdown	Below 18 / 18–25 / 26–30 / 31–35 / 36–40 / Above 40	Yes
Mother's Education	Dropdown	No formal education (can sign) / Primary (up to 5th) / Secondary (6th–10th) /	Yes

Field	Type	Options / Validation	Required
		Higher Secondary / Graduate / Postgraduate+	
Father's Education	Dropdown	Same options as mother	Yes
Mother's Occupation	Dropdown	Homemaker / Govt Service / Private Service / Business / Daily Labour / Other (text)	Yes
Father's Occupation	Dropdown	Unemployed / Govt Service / Private Service / Business / Daily Labour / Other (text)	Yes
Monthly Family Income	Dropdown	Class I: ₹9414+ / Class II: ₹4707–9413 / Class III: ₹2824–4706 / Class IV: ₹1412–2823 / Class V: Below ₹1412	Yes
Type of Family	Radio	Nuclear / Joint / Extended	Yes
Number of Family Members	Dropdown	3 / 5 / Other (number)	Yes
Religion	Dropdown	Hindu / Muslim / Christian / Other (text)	Yes
Area of Residence	Radio	Urban / Rural / Semi-urban	Yes
Contact Number	Tel input	10-digit mobile (may be different from login number)	Optional
Education received about preterm care	Radio	Yes / No	Yes
If yes — source of information	Multi-select	Health workers / Family members / Peer group / Workshop / Magazine	Conditional

A.8 Signup — Step 5: Infant Profile Form

Screen: /signup/baby-profile | Who fills it: Nurse at hospital using NICU records

This screen collects the baby's clinical data. Most of it comes directly from the NICU discharge record — the nurse reads it from the file and enters it. This data is used throughout the app for age calculations, WHO chart comparisons, and vaccination scheduling.

Form Fields — Infant Information

Field	Type	Options / Validation	Used For
Infant's Name / Code	Text	Free text. Code auto-generated separately	Display only
Date of Birth	Date picker	Cannot be future date	Age & corrected age calculation

Field	Type	Options / Validation	Used For
Sex	Radio	Male / Female	WHO chart (different charts per sex)
Gestational Age at Birth	Number	24–36 weeks. Decimals allowed (e.g. 32.5)	Corrected age = chrono age – (40 – GA) weeks
Birth Weight	Number	Grams. 400–4000g range	Baseline growth data
Current Weight at Discharge	Number	Grams	First growth reading
Place of Delivery	Radio	Hospital / Home	Profile data
NICU / SNCU Stay Duration	Number	Days (1–120)	Baseline severity indicator
Skin-to-skin contact at birth (KMC)	Radio	Yes / No	Baseline KMC data
KMC provided in NICU	Radio	Provided / Not Provided	Baseline data
Feeding Type at Discharge	Dropdown	Exclusively breastfed / Exclusively formula / Mixed feeding	Baseline feeding data
Cried after birth	Radio	Yes / No	Baseline clinical data
Need of Resuscitation	Radio	Yes / No	Baseline clinical data

CRITICAL: Gestational Age drives two key calculations: (1) Corrected Age = Chronological Age – (40 – Gestational Age at Birth) weeks — used for WHO growth chart and TDSC. (2) Vaccination schedule — uses CHRONOLOGICAL age (birth date), NOT corrected age. Both must be implemented in backend on account creation.

A.9 Signup — Step 6: Hospital Code Entry

Screen: /signup/hospital-code | Who enters it: Nurse

What the Screen Shows

- Heading: 'Enter Hospital Code'
- Subtext: 'This code was provided by your study coordinator'
- 4–6 character code input field (uppercase)
- 'Confirm' button

Behaviour

Scenario	Behaviour
Valid code entered	Mother's account is linked to that hospital. Hospital name shown: 'Linked to Bankura Medical College'. Proceed.
Invalid code	Show: 'Invalid hospital code. Please check with your nurse.'
Code correct but study enrolment closed	Show: 'Enrolment for this study site is currently closed. Contact Purnima Chakraborty.'

A.10 Signup — Step 7: Participant Code & Group Assignment

Screen: No UI shown to mother — handled by researcher panel

After the hospital code is verified, the system auto-generates a participant code. The group assignment (study group or control group) is determined by the researcher in the admin panel — NOT by the app automatically. This preserves research integrity.

Participant Code Format

Component	Example	Meaning
Hospital prefix	BNK	Bankura Medical College
Group prefix	S or C	S = Study group, C = Control group
Sequential number	042	42nd participant enrolled at this site
Full code	BNK-S-042	Bankura, Study group, participant 42

The researcher assigns the group (S or C) in the admin panel before the nurse starts the signup flow. The app Purnima Chakraborty.

A.11 Signup — Step 8: Confirm Language Preference

Screen: /signup/confirm-language | Shown only during signup

What the Screen Shows

- Heading: 'Choose your preferred language for the app'
- Subtext: 'All daily tips, care guides, and alerts will appear in this language'
- 3 language options (same as A.1) — currently selected one is pre-highlighted
- 'Confirm' button
- Note: 'You can change this later in Settings'

Behaviour

- Language saved to user's profile in the database (preferred_language column)

- All subsequent app content loads in this language
- Language preference syncs across devices if mother re-installs

A.12 Signup Complete — Welcome Screen

Screen: /signup/complete

What the Screen Shows

- Large checkmark icon (green)
- Heading: 'You're all set!' / 'सब किछू ठीकठाक!' / 'सब कुछ तैयार है!'
- Subtext: 'Your account has been created. The app will support you and [baby name] for the next 6 months.'
- Summary card showing: Mother name | Baby name | Hospital | Participant Code | Study Group
- 'Go to Home' button (large, primary)

A.13 Complete Signup Flow Summary

Step	Screen	Who Acts	Can Go Back?
0	Language Selection	Mother / Nurse	Yes — language not saved until next step
1	Welcome Screen	Mother / Nurse	Yes
2	Phone Number Entry	Nurse enters mother's number	Yes
3	OTP Verification	Nurse / Mother	Yes — re-enter number
4	Create 4-digit PIN	Mother (her choice)	No — OTP already used
5	Mother Profile Form	Nurse + Mother	Yes
6	Infant Profile Form	Nurse (from NICU file)	Yes
7	Hospital Code Entry	Nurse	Yes
8	Participant Code Assigned	System auto (no UI)	N/A
9	Confirm Language	Mother	Yes
10	Signup Complete	System	No — account created

Part B — Database Schema

This section defines every table needed in the PostgreSQL database. Four main entities: Hospitals, Users (mothers, researchers, nurses), Babies, and all clinical/research data tables. Follow this schema exactly — changing column names later will break the data export to SPSS.

Database: PostgreSQL | All timestamps stored as UTC | All IDs: UUID v4 | Soft deletes (deleted_at column) on all main tables — never hard delete research data

B.1 Table: hospitals

Stores each NICU/hospital that is a study site. Purnima Chakraborty creates these in the admin panel before enrolment begins.

TABLE: hospitals			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique identifier for this hospital
name	VARCHAR(255)	NOT NULL	Full hospital name e.g. 'Bankura Medical College and Hospital'
code	VARCHAR(10)	NOT NULL, UNIQUE	Short code used during enrolment e.g. 'BNK', 'BWN', 'PRB', 'DBN'
district	VARCHAR(100)	NOT NULL	District name e.g. 'Bankura', 'Burdwan'
state	VARCHAR(100)	NOT NULL, DEFAULT 'West Bengal'	State
type	VARCHAR(50)	NOT NULL	Values: 'primary_site' or 'pilot_site'
emergency_phone	VARCHAR(20)	NULLABLE	NICU helpline number shown on Danger Signs screen
is_active	BOOLEAN	NOT NULL, DEFAULT true	Set false to close enrolment at this site
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Record creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Auto-updated on any change

B.2 Table: users

All system users: mothers, researchers, and nurses. Role column separates them. One table keeps auth simple.

TABLE: users			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY, DEFAULT gen_random_uuid()	Unique user ID
phone	VARCHAR(15)	NOT NULL, UNIQUE	Mobile number with country code e.g. '+919876543210'
pin_hash	VARCHAR(255)	NULLABLE	bcrypt hash of 4-digit PIN. NULL for researchers (they use password)
password_hash	VARCHAR(255)	NULLABLE	bcrypt hash. Only for role = researcher or nurse
role	VARCHAR(20)	NOT NULL	Values: 'mother', 'researcher', 'nurse'
preferred_language	VARCHAR(5)	NOT NULL, DEFAULT 'bn'	Values: 'bn' (Bengali), 'hi' (Hindi), 'en' (English)
hospital_id	UUID	FK → hospitals.id, NULLABLE	NULL for researcher. Set for nurse and mother.
is_active	BOOLEAN	NOT NULL, DEFAULT true	False = account suspended or study ended
last_login_at	TIMESTAMPTZ	NULLABLE	Updated on every successful login
failed_pin_attempts	INTEGER	NOT NULL, DEFAULT 0	Reset to 0 on successful login. Lock at 5.
locked_until	TIMESTAMPTZ	NULLABLE	Account locked until this time after too many failures
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Account creation timestamp
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Auto-updated
deleted_at	TIMESTAMPTZ	NULLABLE	Soft delete — set when account closed. Never hard delete.

B.3 Table: otp_verifications

Stores OTP codes sent during signup and PIN reset. Expired OTPs are never deleted — kept for audit trail.

TABLE: otp_verifications			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	Unique OTP record ID
phone	VARCHAR(15)	NOT NULL	Phone number OTP was sent to
otp_hash	VARCHAR(255)	NOT NULL	SHA-256 hash of the 6-digit OTP — never store plain OTP

TABLE: otp_verifications			
purpose	VARCHAR(30)	NOT NULL	Values: 'signup', 'pin_reset', 'login_fallback'
is_used	BOOLEAN	NOT NULL, DEFAULT false	Set true after successful verification
attempts	INTEGER	NOT NULL, DEFAULT 0	Number of wrong attempts on this OTP
expires_at	TIMESTAMPTZ	NOT NULL	OTP valid for 10 minutes from creation
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	When OTP was generated
used_at	TIMESTAMPTZ	NULLABLE	When OTP was successfully used

B.4 Table: refresh_tokens

TABLE: refresh_tokens			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	Token ID
user_id	UUID	NOT NULL, FK → users.id	Which user this token belongs to
token_hash	VARCHAR(255)	NOT NULL	SHA-256 hash of the token value
device_info	TEXT	NULLABLE	Browser / device info for security display
expires_at	TIMESTAMPTZ	NOT NULL	30 days from creation
revoked_at	TIMESTAMPTZ	NULLABLE	Set when user logs out
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Token creation time

B.5 Table: mother_profiles

All demographic data for Tool I of the research proposal — mother section. One row per mother.

TABLE: mother_profiles			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	Profile ID
user_id	UUID	NOT NULL, UNIQUE, FK → users.id	Links to users table. UNIQUE = one profile per user

TABLE: mother_profiles			
participant_code	VARCHAR(20)	NOT NULL, UNIQUE	e.g. 'BNK-S-042'. Used in all research data exports
study_group	VARCHAR(10)	NOT NULL	Values: 'study' or 'control'
hospital_id	UUID	NOT NULL, FK → hospitals.id	Which hospital this mother was enrolled at
full_name	VARCHAR(255)	NULLABLE	Optional. Not included in anonymised exports.
age_range	VARCHAR(30)	NOT NULL	e.g. '18-25', '26-30' — from dropdown
education_mother	VARCHAR(50)	NOT NULL	Education level from Tool I options
education_father	VARCHAR(50)	NOT NULL	Education level from Tool I options
occupation_mother	VARCHAR(50)	NOT NULL	Occupation from Tool I options
occupation_father	VARCHAR(50)	NOT NULL	Occupation from Tool I options
income_class	VARCHAR(20)	NOT NULL	BG Prasad scale: 'I','II','III','IV','V'
family_type	VARCHAR(20)	NOT NULL	Values: 'nuclear', 'joint', 'extended'
family_members_count	VARCHAR(10)	NOT NULL	Values: '3', '5', 'other' + number
religion	VARCHAR(30)	NOT NULL	Hindu / Muslim / Christian / Other
residence_type	VARCHAR(20)	NOT NULL	Values: 'urban', 'rural', 'semi_urban'
contact_number	VARCHAR(15)	NULLABLE	Optional alternate contact
prev_preterm_education	BOOLEAN	NOT NULL, DEFAULT false	Whether mother received prior preterm care education
education_source	TEXT[]	NULLABLE	Array: ['health_worker', 'family', 'peer', 'workshop', 'magazine']
enrolled_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	Date of enrolment / account creation
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	

B.6 Table: baby_profiles

All infant demographic and clinical data from Tool I — infant section. One row per baby. All age calculations in the app derive from this table.

TABLE: baby_profiles			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	Baby profile ID
mother_profile_id	UUID	NOT NULL, UNIQUE, FK → mother_profiles.id	Links baby to mother. UNIQUE = one baby per mother (study constraint)
baby_name	VARCHAR(100)	NULLABLE	Optional display name
sex	VARCHAR(10)	NOT NULL	Values: 'male', 'female'
date_of_birth	DATE	NOT NULL	Used for chronological age and vaccination schedule
gestational_age_weeks	DECIMAL(4,1)	NOT NULL	e.g. 32.5 — used for corrected age calculation
birth_weight_grams	INTEGER	NOT NULL	Birth weight in grams — baseline growth data point
weight_at_discharge_grams	INTEGER	NOT NULL	Weight when leaving NICU — first growth tracking point
place_of_delivery	VARCHAR(20)	NOT NULL	Values: 'hospital', 'home'
nicu_stay_days	INTEGER	NOT NULL	Number of days in NICU
skin_to_skin_at_birth	BOOLEAN	NOT NULL	KMC immediately at birth — Yes/No
kmc_in_nicu	BOOLEAN	NOT NULL	Was KMC provided during NICU stay — Yes/No
feeding_at_discharge	VARCHAR(30)	NOT NULL	Values: 'exclusive_bf', 'exclusive_formula', 'mixed'
cried_at_birth	BOOLEAN	NOT NULL	Cried after birth — Yes/No
needed_resuscitation	BOOLEAN	NOT NULL	Required resuscitation — Yes/No
birth_weight_stratum	VARCHAR(20)	NOT NULL	Auto-calculated on save: 'under_1500', '1500_to_2500', 'over_2500'. Used for randomisation stratification.
discharge_date	DATE	NOT NULL	NICU discharge date — study start date for this participant
created_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	
updated_at	TIMESTAMPTZ	NOT NULL, DEFAULT NOW()	

Computed values (not stored — calculated on query):• Chronological age = TODAY – date_of_birth• Corrected age = Chronological age – (40 – gestational_age_weeks) weeks• Corrected age is used for: WHO growth chart, TDSC screening• Chronological age is used for: Vaccination schedule

B.7 Table: nurse_profiles

Nurses are hospital staff who handle enrolment. They have limited access — enrolment flow only.

TABLE: nurse_profiles			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	
user_id	UUID	NOT NULL, UNIQUE, FK → users.id	Links to users table
hospital_id	UUID	NOT NULL, FK → hospitals.id	Nurse belongs to this hospital
full_name	VARCHAR(255)	NOT NULL	Nurse's name
employee_id	VARCHAR(50)	NULLABLE	Hospital employee ID (optional)
is_active	BOOLEAN	NOT NULL, DEFAULT true	Set false when nurse leaves
created_at	TIMESTAMP TZ	NOT NULL, DEFAULT NOW()	

B.8 Table: researcher_profiles

Purnima Chakraborty and any research assistants she designates. Full access to admin panel.

TABLE: researcher_profiles			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	
user_id	UUID	NOT NULL, UNIQUE, FK → users.id	Links to users table
full_name	VARCHAR(255)	NOT NULL	Researcher's full name
designation	VARCHAR(100)	NULLABLE	e.g. 'Principal Investigator', 'Research Assistant'
email	VARCHAR(255)	NOT NULL, UNIQUE	Login email for researcher (they use email+password not phone+PIN)
access_level	VARCHAR(20)	NOT NULL, DEFAULT 'full'	Values: 'full' (Purnima Chakraborty, 'read_only' (external reviewer)
created_at	TIMESTAMP TZ	NOT NULL, DEFAULT NOW()	

B.9 Table: follow_up_schedules

Auto-generated when a mother is enrolled. Creates 4 follow-up records (baseline + 3 months). Used to track whether data has been collected at each time point.

TABLE: follow_up_schedules			
Column Name	Data Type	Constraints	Description / Notes
id	UUID	PRIMARY KEY	
mother_profile_id	UUID	NOT NULL, FK → mother_profiles.id	Which participant
time_point	VARCHAR(20)	NOT NULL	Values: 'baseline', '1_month', '3_months', '6_months'
scheduled_date	DATE	NOT NULL	Auto-calculated from discharge_date + interval
actual_date	DATE	NULLABLE	Date data was actually collected (may differ from scheduled)
status	VARCHAR(20)	NOT NULL, DEFAULT 'pending'	Values: 'pending', 'completed', 'missed', 'rescheduled'
data_complete	BOOLEAN	NOT NULL, DEFAULT false	True only when all 5 tools submitted for this time point
collected_by_user_id	UUID	NULLABLE, FK → users.id	Researcher or nurse who collected data
notes	TEXT	NULLABLE	Any notes from researcher about this visit
created_at	TIMESTAMP TZ	NOT NULL, DEFAULT NOW()	
updated_at	TIMESTAMP TZ	NOT NULL, DEFAULT NOW()	

B.10 Entity Relationship Summary

How all tables connect to each other:

Table	Connects To	Relationship	Purpose
hospitals	users, mother_profiles, nurse_profiles	One hospital → many users	Sites management
users	mother_profiles, nurse_profiles, researcher_profiles	One user → one profile	Auth for all roles
mother_profiles	baby_profiles, follow_up_schedules	One mother → one baby, one → many follow-ups	Core research entity
baby_profiles	mother_profiles	One baby → one mother	Infant clinical data
nurse_profiles	users, hospitals	One nurse → one hospital	Enrolment staff
researcher_profiles	users	One researcher → one user	Admin panel access
follow_up_schedules	mother_profiles	One mother → 4 schedules	Research timeline tracking
otp_verifications	users (via phone)	Many OTPs per phone	Auth security
refresh_tokens	users	Many tokens per user	Session management

Part C — What to Build First (Developer Checklist)

Start here. Everything else in the app depends on these tables and screens being correct. Get this foundation right and the rest of the project builds cleanly on top of it.

Backend — Build in This Order

11. **Create PostgreSQL database with all 9 tables from Part B**
12. Set up UUID generation, timestamps, and soft delete triggers
13. POST /auth/send-otp — accepts phone number, sends OTP via MSG91, stores hashed OTP in otp_verifications
14. POST /auth/verify-otp — verifies OTP hash, marks as used, returns a short-lived signup token
15. POST /auth/create-pin — accepts signup token + PIN, creates user in users table with bcrypt PIN hash
16. POST /auth/login — accepts phone + PIN, returns JWT access token + refresh token
17. POST /auth/refresh — accepts refresh token, returns new access token
18. POST /auth/logout — revokes refresh token
19. POST /auth/forgot-pin — sends OTP for PIN reset
20. POST /onboarding/mother-profile — saves all mother demographic fields to mother_profiles
21. POST /onboarding/baby-profile — saves all infant fields to baby_profiles, auto-calculates birth_weight_stratum
22. POST /onboarding/hospital-code — validates hospital code, links user to hospital
23. GET /onboarding/participant-code — returns next available participant code for this hospital + group
24. POST /onboarding/complete — marks onboarding done, generates 4 follow_up_schedule records

Frontend — Build in This Order

25. **Language selection screen (A.1) with i18n setup — Bengali, Hindi, English**
26. Welcome screen (A.2)
27. Phone number entry screen (A.3) with validation
28. OTP verification screen (A.4) with 6-box input and countdown timer
29. 4-digit PIN creation screen (A.5) with confirmation
30. Mother profile form (A.7) — all demographic dropdowns and radios in Bengali
31. Baby profile form (A.8) — all infant fields with NICU data entry
32. Hospital code entry screen (A.9)
33. Signup complete screen (A.12)
34. Returning login screen (A.6) — phone + PIN
35. Forgot PIN flow (OTP → new PIN)

Testing Checklist Before Moving to Next Phase

- All 9 database tables created with correct data types and foreign keys
- OTP is sent via SMS and received on a real Indian phone number
- OTP expiry (10 min) and attempt limiting (3 tries) working correctly
- PIN is hashed — never stored in plain text — verified in database directly

- JWT access token expires in 24 hours — refresh token in 30 days
- A mother can complete full signup in Bengali from language select to dashboard
- Participant code is generated in correct format (e.g. BNK-S-042)
- birth_weight_stratum is auto-calculated correctly on baby profile save
- follow_up_schedule creates 4 records with correct dates on signup complete
- A returning mother can log back in with phone + PIN
- Forgot PIN flow works end to end
- All form validation errors appear in Bengali